

Milestone 2 — Speaker Companion

15-minute presentation: per-slide guidance, background, and Q&A backup

Pacing. 13 main-content slides over ~15 minutes $\approx$ 45–75 seconds per slide. Spend most time on the highlighted headline-result and in-space placebo slides; move quickly through the methodology recap. Appendix slides are reserve material for Q&A — do not budget time for them. **Watch the clock at slide 11 (headline)**; if behind, compress Part 2 (slide 13) and go straight to thanks.

1. Slide 1 — Title

[10s]

Open with the title. The framing is causal estimation of one specific event, not forecasting and not the network-topology project from the previous milestone. Mention that today is Part 1 of a two-part thesis.

2. Slide 2 — From the previous presentation

[45s]

Talking points

- **Original framing** (as on the slide): *How does the topology of the global maritime oil trade network shape the dynamics of oil prices?*
 - **Q1:** Can network topology improve oil-price forecasts beyond classical time-series baselines?
 - **Q2:** How does a maritime disruption propagate into oil prices across horizons, and how long does its effect linger?
- **Why the pivot to causal SCM:**
 - Focus on *causal* estimation rather than forecasting — forecasting is crowded and hard to falsify; a credible counterfactual on one specific disruption is more tractable and policy-relevant.
 - Reviewed candidate causal methods: event studies, DiD, network-shock propagation, synthetic control.
 - SCM selected for the combination of *inferential machinery* (in-space placebo) and *interpretable result* (donor-weighted counterfactual on the observed price), plus it produces an ATT directly rather than a shock decomposition.

Behind the slide

The maritime-network framing was scoped down rather than abandoned. Event studies don't produce a continuous counterfactual; DiD needs a clear treated/control structure that doesn't exist for a single global commodity; network-shock propagation would require strong identification assumptions on the network itself. Structural VAR (Kilian-style) is mentioned in the Lit-overview slide as the field default but was ruled out here because its identification restrictions are contested for chokepoint events.

If asked

- *Why not just stick with the network framing?* Network propagation on oil prices is interesting but very hard to identify from observational data; the network-as-donor approach we considered ends up looking similar to SCM with engineered features. SCM gives a cleaner identification story now.
- *Could the network framing come back in Part 2?* Yes — network centrality or routing-disruption features could enter the donor pool, or be layered as a propagation model on top of the ATT. Part 2 is open.

- *Why drop Local Projection / structural VAR from the candidate list?* LP and SVAR appear on the Lit-overview slide (slide 4) as the field-default approaches we are positioned against. They're not in the candidate list on slide 2 because they're not what we ultimately chose; we mention them on slide 4 to show we're aware of the literature.

3. Slide 3 — Research question (Part 1)

[60s]

Talking points

- **The question (in the red box):** *What was the causal effect of the 2026 Strait of Hormuz disruption on the Brent crude price?*
- **Estimand:** average treatment effect on the treated (ATT) over a defined post-event window. One treated unit (Brent) → effect *for that unit*, not a population average.
- **Approach (left column):** SCM on the observed Brent price series. Ensemble of five model specifications to span the SCM regularization design space.
- **Answer today (right column):** Brent traded at a **~44%** premium above the donor-weighted counterfactual over the March–May 2026 post-event window. IQR across the five models: [40%, 44%].
- **Part 2** is open — come back to it on slide 15.

Behind the slide

This is the slide where you state the punchline up front: the headline number (~44%) and the IQR. The audience then knows what to evaluate as you walk through the methodology slides. Don't yet defend the number — just say it.

The Part 2 placeholder is intentional — it tells the audience the second half of the thesis is open to direction, which softens the methodology focus of Part 1.

If asked

- *Why ATT and not ATE?* One treated unit (Brent). There is no population to average over; the estimand is the effect on the realized series.
- *Why is there no peer-reviewed SCM precedent on crude?* Combination of recency (price-impact SCM is mostly post-2020), data availability (good cross-asset donor pools for commodities are non-obvious), and convention (the oil literature defaults to structural VAR; SCM is more common in policy-evaluation settings). Closest mechanical precedent is Becker et al. 2021 on regulated Austrian retail fuel, covered on slide 4.
- *What's the policy use of ~44%?* SPR sizing, central-bank reaction function (oil shocks pass through to headline inflation), fiscal buffers, sovereign hedging.

4. Slide 4 — How the field estimates oil-price impacts

[30s]

Talking points

- Three field-default approaches, briefly: **Structural VAR** (Kilian 2009; identification restrictions contested for chokepoint events, no continuous counterfactual path); **Event study** (no path beyond a few days, persistence not captured); **Scenario models** (Dallas Fed, OIES, Kiel, IEA; counterfactual is assumed, not read from observed prices).
- Two SCM-on-price precedents: **Becker, Pfeifer & Schweikert 2021** on Austrian retail fuel (closest mechanical design); **Gharehgozli 2017** on Iran GDP under sanctions (establishes SCM for geopolitical disruptions but on GDP, not market price).
- **Gap we fill (red box):** no published study applies SCM to a globally traded crude benchmark under an exogenous geopolitical disruption.

Behind the slide

Move quickly through this slide — it’s positioning, not the contribution itself. The point is to show we know the landscape and to set up slide 5 where we say what we do differently.

5. Slide 5 — What we do differently

[45s]

Talking points

- **Three design extensions** vs the Becker et al. precedent:
 - Retail fuel → *crude benchmark*. Brent underpins retail fuel, CPI, and sovereign hedging.
 - Cross-country → *cross-asset donors*. No crude benchmark is Hormuz-clean (every crude benchmark inherits the shock), so donors must come from metals, FX, equities, rates.
 - Regulatory → *geopolitical* treatment. Treatment is exogenous and unannounced; SUTVA reasoning is about asset-class routing, not policy endogeneity.
- **Two-event validation design** (own contribution; no canonical SCM precedent): Hormuz 2026 is the focal estimate, Russia 2022 is run as a magnitude validation. SCM Russia counterfactual lands within \$3.98/bbl of EIA STEO’s last pre-invasion structural forecast on the matched 151-day window (~5.7 pp on implied ATT).
- **Complement, not substitute**: scenario models ask “what price does an *assumed* shortfall imply?”; SCM asks “what gap does *observed market data* imply?” Agreement → independent validation. Disagreement → reveals what structural assumptions miss.
- **Additional value** SCM provides over the field defaults: full price path (integrates persistence), auditable identification (donor pool open to challenge), no structural stance (total reduced-form effect, no prior on supply vs. risk premium split).

Behind the slide

This is where the contribution lives. Spend the most time on the two-event design — it’s the methodological novelty and the easiest target for committee questions.

If asked

- *Why is Russia validation credible if it has macro confounders?* Weakened but not invalidated — detailed defense on slide 11 (model fit) and in the “Hardest critiques” section below.

6. Slide 6 — Brent spot price, post-COVID, curated event timeline

[45s]

Talking points

- Visual orientation: where Hormuz 2026 sits in the broader post-COVID Brent path. Roughly \$60/bbl in 2020, runs up through 2021–22 (Russia), declines through 2023–24, then the Hormuz spike in early 2026.
- Events colour-coded by category (demand, supply, policy, geopolitics) — shows the data is event-rich, which is why donor cleanliness matters.
- Highlight the two focal events visually: **Hormuz 2026** (focal) and **Russia 2022** (calibration).

If asked

- *Why these two events specifically?* Both are large, well-measured, recent geopolitical disruptions. Hormuz 2026 is the focal event of interest; Russia 2022 is the most recent large oil shock with a long enough post-window to be well-measured, used as a magnitude-validation case.
- *Why not Abqaiq 2019 or others?* Discussed in donor_catalog.md: pre-windows for older events overlap with Hormuz’s, breaking the cross-event independence we exploit in §5f. Russia 2022 is independent of the Hormuz pre-window.

Talking points

- Ensemble of 5 models spanning the SCM design space:
 - **Convex SCM** (Abadie, Diamond & Hainmueller 2010): canonical, no extrapolation.
 - **Augmented SCM** (Ben-Michael, Feller & Rothstein 2021): ridge bias correction to allow extrapolation when convex hull is too tight.
 - **Elastic-net** (Doudchenko & Imbens 2016): $L_1 + L_2$ regression, sparse signed weights.
 - **XGBoost** (Chen & Guestrin 2016): nonlinear, captures interactions.
 - **Bayesian Ridge**: Bayesian linear regression with normal prior. Note: this is NOT the full state-space BSTS of Brodersen 2015, just the regression component (clarified in methodology.md).
- **Why an ensemble**: the headline is the median across models; IQR is the model-uncertainty band. No single model assumption is privileged.

Behind the slide

The ensemble spans key SCM regularization assumptions: simplex (convex SCM), ridge (ASCM, BR), L_1+L_2 (Elastic-net), tree-based (XGBoost). Disagreement across models is itself informative: tight agreement = factor structure is well-identified; wide spread = model-uncertainty band that the headline IQR captures honestly.

Model formulas — what each one fits

Let $y_t = \log\text{-Brent}$ at time t and $y_{j,t} = \log\text{-price}$ of donor j . All models produce a synthetic counterfactual $\hat{y}_t$; the post-event gap is $100 \cdot (\exp(\bar{y}_{post}) - \bar{\hat{y}}_{post}) - 1$ in percent.

- **Convex SCM**. Synthetic is a *simplex-constrained weighted average* of donors:

$$\hat{y}_t = \sum_{j=1}^J w_j y_{j,t}, \quad w_j \geq 0, \quad \sum_j w_j = 1.$$

Read like portfolio shares: “synthetic Brent is w_A of donor A + w_B of donor B + ...”. Cannot go outside the donor envelope. Sparse: typically ~ 5 donors get non-trivial weight, the rest pinned to zero. *Feature selection*: the simplex constraint behaves like an L_1 ball — the SLSQP optimizer drives most w_j to exactly zero, so “best features” = the small set of donors whose convex combination best reproduces Brent in the pre-period (lowest pre-RMSPE). No external selection step; sparsity is mechanical.

- **Augmented SCM**. Convex SCM constrains the donor weights to be non-negative and sum to 1 — interpretable but rigid. ASCM keeps those convex weights and adds a second ridge-regression layer that *extrapolates the donors’ contribution beyond the simplex* — letting the synthetic borrow more (or less) from each donor than Convex SCM allowed:

$$\hat{y}_t = \underbrace{\sum_j w_j y_{j,t}}_{\text{convex base}} + \underbrace{\sum_j \eta_j y_{j,t}}_{\text{ridge extrapolation}}, \quad \eta = (X^\top X + \lambda I)^{-1} X^\top r,$$

where r is the pre-period gap left over by convex SCM and X is the donor matrix. The effective weight per donor is $w_j + \eta_j$ — the η_j are unconstrained (can be negative or exceed 1), which is what produces the extrapolation outside the donor hull. The hyperparameter $\lambda = \text{ridge_lambda}$ controls how much extrapolation is permitted: large λ shrinks the ridge layer toward zero and recovers Convex SCM; small λ gives the ridge layer free rein. *Feature selection*: sparsity is inherited from the convex base (w_j) only; the ridge layer η_j touches every donor (no selection). So “best features” = same as Convex SCM, plus every donor contributing a small ridge correction.

- **Elastic-net.** Linear regression with combined $L_1 + L_2$ penalty:

$$\hat{y}_t = \beta_0 + \sum_j \beta_j y_{j,t}, \quad \min_{\beta} \frac{1}{2N} \|y - X\beta\|_2^2 + \alpha \left[\rho \|\beta\|_1 + \frac{1-\rho}{2} \|\beta\|_2^2 \right].$$

Coefficients β_j are signed and unconstrained; the L_1 term drives some to exactly zero (sparsity), the L_2 term shrinks the rest. α = overall penalty strength; $\rho = \text{11_ratio}$ (0 = pure ridge, 1 = pure lasso, 0.5 = balanced). *Feature selection:* explicit via the L_1 term — donors with weak partial correlation against Brent’s residual get $\beta_j = 0$. Higher $\rho \rightarrow$ sparser selection. “Best features” = donors whose marginal contribution survives the L1 threshold after accounting for collinearity with already-selected donors.

- **XGBoost.** Sum of K shallow regression trees, fit sequentially to residuals:

$$\hat{y}_t = \sum_{k=1}^K f_k(\mathbf{y}_{\cdot,t}), \quad f_k \in \{\text{trees of depth} \leq d\},$$

where each tree f_k minimizes the residual after the previous $k - 1$ trees, with L_2 regularization λ on leaf weights and step size η (learning rate). *Not* a closed-form regression: a tree at each step picks the donor and threshold that best reduces the residual. Captures nonlinear interactions but cannot extrapolate outside the training range of any donor. *Feature selection:* greedy and built-in. At every split, the tree-builder scores every donor’s best threshold by gain reduction and picks the maximum; donors that never get picked have zero importance. “Best features” = donors with highest cumulative split-gain across all K trees (reported as XGBoost’s `feature_importances_`).

- **Bayesian Ridge.** Same regression equation as OLS, but with a Gaussian prior placed on the coefficients β (and Gamma priors on the precisions):

$$\underbrace{y_t = \beta_0 + \sum_j \beta_j y_{j,t} + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, 1/\alpha)}_{\text{data model (identical to OLS)}}$$

$$\underbrace{\beta_j \mid \lambda \sim \mathcal{N}(0, 1/\lambda), \quad \lambda \sim \text{Gamma}(\cdot), \quad \alpha \sim \text{Gamma}(\cdot)}_{\text{prior on the unknowns (what makes it Bayesian)}}$$

The features $y_{j,t}$ enter the same way they would in OLS — as known data inside the regression sum. The Bayesian part is purely the prior on the unknowns (β , α , λ). Posterior over β is closed-form (conjugate), so $\hat{y}_t$ comes with a native posterior standard deviation, no bootstrap needed. λ (the weight-precision prior) is the regularization knob — larger $\lambda \rightarrow$ tighter prior around zero $\rightarrow$ stronger shrinkage; α is the noise-precision prior. MAP-equivalent to classical Ridge with penalty λ/α . *Feature selection: none — all donors retained.* The Gaussian prior shrinks coefficients smoothly toward zero but never sets any to exactly zero, unlike L_1 models. “Best features” is approximated by the PIP proxy $|\hat{\beta}_j|/\sigma_{\hat{\beta}_j}$ (the posterior z -score), which ranks donors by how confidently their coefficient differs from zero. To get *actual* variable selection in a Bayesian framework, you would need spike-and-slab priors (the full BSTS of Brodersen 2015), which this implementation does not include.

Tuned hyperparameters — chosen by walk-forward CV val RMSE

Grids tightened per Cawley & Talbot (2010) to limit winner’s-curse on val RMSE. Values below are the *chosen* settings per event after the 5-fold walk-forward search.

- **Convex SCM** — no tuned hparams. Fixed: 120 random V matrices drawn from a Dirichlet, weights re-fit per V , best by pre-period RMSPE retained. Same for both events.
- **Augmented SCM** (grid: $\lambda \in \{0.01, 0.1, 1, 10, 100\}$, 5 candidates):
 - Hormuz: `ridge_lambda` = 10.0 (heavier extrapolation control).

- Russia: `ridge_lambda = 0.1` (light extrapolation control — the convex residual is bigger so ridge needs more freedom).
- **Elastic-net** (grid: $\alpha \in \{0.001, 0.01, 0.05, 0.1, 0.5\} \times \rho \in \{0.2, 0.5, 0.8\}$, 15 candidates):
 - Hormuz: `alpha = 0.001, l1_ratio = 0.8` (sparser, lasso-leaning).
 - Russia: `alpha = 0.001, l1_ratio = 0.2` (denser, ridge-leaning).
- **XGBoost** (grid: `max_depth` $\in \{2, 3, 4\} \times$ `learning_rate` $\in \{0.001, 0.005, 0.01, 0.03, 0.1\} \times$ `reg_lambda` $\in \{1, 10\}$, 30 candidates; `n_estimators=1000` with early stopping after 20 rounds):
 - Hormuz: `max_depth=2, learning_rate=0.1, reg_lambda=10.0`.
 - Russia: `max_depth=2, learning_rate=0.1, reg_lambda=1.0`.

Both events stay at depth 2 — only second-order interactions, consistent with the small-data regime.

- **Bayesian Ridge** (grid: $\lambda \in \{10^{-8}, 10^{-6}, 10^{-4}, 10^{-2}, 1\} \times \alpha \in \{10^{-8}, 10^{-6}, 10^{-4}\}$, 15 candidates):
 - Hormuz: $\lambda_1 = \lambda_2 = 1.0, \alpha_1 = \alpha_2 = 10^{-8}$ (strong weight shrinkage, near-flat noise prior).
 - Russia: $\lambda_1 = \lambda_2 = 1.0, \alpha_1 = \alpha_2 = 10^{-8}$ (same).

Both events converge on the strongest prior in the grid — the donor regression is high-variance at $N \approx 420$ and shrinkage helps both.

If asked

- *Why not include BSTS proper?* The implementation in `lib/models.py` is `sklearn.linear_model.BayesianRidge` — Bayesian linear regression on donor levels, not the full state-space BSTS with trend + seasonal + spike-and-slab. Documented honestly in `methodology.md` §4 “Bayesian Ridge vs full BSTS.”
- *Why not include simpler models like OLS?* OLS without regularization overfits at $N \approx 420$ obs and 21 donors. Elastic-net at low α approximates OLS while preventing overfit.
- *Why does ASCM pick very different λ across the two events?* Russia has a larger convex-residual to absorb (Brent’s pre-period $RMSPE \approx 0.106$ vs Hormuz’s ~ 0.066), so the ridge fix-up needs more freedom (lower λ) to fit it. Hormuz’s convex fit is tight enough that a heavy λ doesn’t hurt and protects against extrapolation.
- *Why does Elastic-net pick opposite `l1_ratio` across events?* Hormuz’s signal is concentrated in a few donors (sparser fit preferred $\rightarrow \rho=0.8$); Russia’s signal is spread across many correlated donors (ridge-leaning $\rightarrow \rho=0.2$).

8. Slide 8 — Model training and validation

[75s]

Talking points

- Pre-event windows: Hormuz 2024-06 $\rightarrow$ 2026-01 (423 obs); Russia 2020-07 $\rightarrow$ 2022-02 (421 obs). Post-event: ~ 59 obs Hormuz / 151 obs Russia.
- **Training:** 5-fold expanding-window walk-forward CV (Hyndman & Athanasopoulos 2018; Bergmeir & Benítez 2012). Hyperparameters chosen by pooled val RMSE across folds; final fit refits on full pre-event window.
- Validation battery: model fit (walk-forward), SUTVA (donor audit), inference (in-space placebo + cross-event transfer).

Behind the slide

The walk-forward CV is a recent refactor (previously was a single 80/20 hold-out, which is a time-series train-test split, not true walk-forward CV per Hyndman). The 5-fold expanding-window scheme is more robust because it averages over 5 forecast windows rather than concentrating all val mass on the

last 20%. Headline numbers shifted modestly under the refactor: Hormuz unchanged at 43%; Russia ensemble median 25.1% → 21.7%.

If asked

- *Why expanding-window rather than sliding-window or pure k-fold?* K-fold breaks time-series structure (uses future to predict past). Sliding-window discards early data. Expanding-window matches the operational reality (you have more data as you go).
- *Hyperparameter selection bias?* Yes: hparams are picked to minimize val RMSE, so reported val RMSE is biased downward. Documented in validation.md §5a as a “winner’s-curse” caveat. Mitigation: tightened grids (each model has at most 9 candidates) per Cawley & Talbot 2010.
- *Why not a separate test set?* Pre-event window is sample-size-constrained. Splitting off a test set would shrink training meaningfully. Formal inference is delegated to the in-space placebo (which is genuinely out-of-sample at the unit level).

9. Slide 9 — Donor selection

[75s]

Talking points

- **21 audited donors out of 33 candidates.** The full 33-donor catalog was audited for both pre-treatment co-movement with Brent AND treatment cleanliness (SUTVA). 12 dropped due to event-specific exposure.
- Categories: metals (3), agriculturals (4), equities (3), FX (8), rates/credit (2), volatility (1).
- Inclusion rule: pre-treatment co-movement through a *common global factor* AND treatment cleanliness (no Russia/Ukraine or Persian Gulf exposure).
- Same pool used for both events, enabling the cross-event weight transfer test (§5f).

Behind the slide

Donor selection is the SCM-canonical SUTVA defense (Abadie 2021 §3.1: substantive grounds, not statistical tests). The audit reasoning is in donor_catalog.md. The dropped 12 includes: Russia commodities (Wheat, Palladium, Russian-channel FX like EUR), risk-off proxies that loaded on Russia specifically (DXY, BTC due to sanctions-evasion narrative), and others.

If asked

- *What’s the full list of dropped donors?* See backup slide “Excluded donors.” 12 Russia-exposed (H or M tier): Copper, IronOre, Palladium, Soybeans, Wheat, Corn, EM_Eq, EUR, GBP, DXY, US10Y, BTC. Plus another set of categorical exclusions (energy complex itself: WTI, NatGas, refined products; oil-currency: NOK, CAD; etc.) that never entered the candidate pool.
- *What if a clean donor is actually treated?* The model-agnostic battery in 01.5_Donor_Cleanliness checks this empirically. Hormuz: 32/33 agreement between audit and tests; Russia: 20/33. Discrepancies are concurrent macro events (CNY’s China zero-COVID), not Russia-treatment.
- *Why 21 specifically and not 24 or 18?* The shared pool is the intersection of Russia-clean (21 of 33) and Hormuz-clean (33 of 33). Russia is the binding constraint.

10. Slide 10 — Donor co-movement with Brent (audited pool)

[30s]

Talking points

- Visual: 21 audited donors rebased to 100 at pre-window start, with Brent in black on top. Dashed line marks T_0 .
- Show the audited donors are visibly correlated with Brent in the pre-event window — the SCM has something real to fit.
- Move quickly; this is an orientation slide, not an analytical one.

Talking points

- Ensemble paths (left Hormuz, right Russia): all 5 model synthetics overlay observed Brent in the pre-event window, then fan out post-event. Visual evidence of pre-event fit AND post-event divergence.
- Walk-forward CV table: **train RMSE / val RMSE / val-train ratio per model, both events**. Numbers are in **log-Brent units** (the fit target is log-price levels, not log-returns).
- Convex SCM val/train ratio is 1.26 Hormuz / 1.16 Russia (clean). On Russia, ASCM 2.04 (just over $2\times$ heuristic threshold), XGBoost 3.00, and Bayesian Ridge 3.07 are the overfit-flagged cases.
- Hormuz: 4 of 5 models clean; only Bayesian Ridge flagged.
- Ensemble is robust to individual model overfit because median across 5 models smooths.

Hormuz vs Russia — the headline distinction

Hormuz is the focal causal estimate. Clean post-period with no concurrent macro confounder; all 5 models converge in the 38–50% range; in-space placebo hits the floor for every cell. Identification holds.

Russia is the methodology-validation case. Identification is partially weakened because the post-window (Feb–Sep 2022) overlaps a major macro regime shift:

- Fed liftoff March 2022 (+25 bp), then +75 bp $\times$ 3 consecutive hikes.
- DXY surged $\sim 95 \rightarrow \sim 115$.
- JPY collapsed $\sim 115 \rightarrow \sim 145$ (BoJ stayed on YCC, carry trade unwound).
- Bond crash (TLT -30%), global risk-off, credit-spread widening.

These hit the donor pool (JPY, KRW, MXN, TLT, HYG, WorldEq, ...) even though the qualitative audit certified them “Russia-clean.” They moved for *macro* reasons, not Russia-spillover.

The cleanest read of the Russia premium is in the first 2–3 weeks post-invasion: Brent went \$95 (Feb 23) $\rightarrow$ \$130+ (Mar 8), well before Fed hiking became dominant. A shorter post-window would isolate the Russia effect more cleanly — one-line config change in `lib/config.py` if a reviewer asks.

Defense for the Russia case in spite of weakened identification: (a) EIA STEO agreement within \$3.98/bbl on the matched window, (b) sparse-model convergence around 22–30% (which exclude the macro-confounded donors via L_1 /simplex sparsity), (c) well-documented historical magnitude. The Russia ATT is *not* a clean point estimate; it’s a credibility check that the methodology recovers a believable magnitude on a known disruption.

Why Bayesian Ridge’s Russia synthetic “always increases”

The visible failure mode: BR’s Russia synthetic tracks Brent’s post-event rally rather than staying flat, eating the Russia premium and giving a degenerate +0.5% ATT. Three stacked reasons:

- **Dense regression, no feature selection.** BR retains all 21 donors with non-zero coefficients (unlike Convex SCM / Elastic-net, which zero out most). When the macro-confounded donors moved up post-event, BR’s synthetic moved up with them.
- **Russia’s macro confounder pushes those donors in the same direction as Brent.** Fed hiking $\rightarrow$ dollar surge $\rightarrow$ JPY collapse, bond crash, equity drawdown. Dense weights pick all that up; sparse weights drop most of it.
- **BR is overfit on Russia** (val/train ratio 3.07). Overfit models amplify input-feature movements at test time.

Hormuz post-period has no comparable macro shock, so BR’s Hormuz synthetic stays flat alongside the other models and the ATT is in line (+43.6%).

Interpreting the RMSE numbers — log-Brent to dollars

The fit target is log-Brent (not log-returns), so RMSE is in log-price units. Conversion: a residual σ in log-units $\rightarrow$ multiplicative error e^σ ; for small σ , $e^\sigma - 1 \approx \sigma$, so **log-RMSE $\approx$ fractional error to first order**.

Log RMSE	$e^\sigma - 1$	% typical err.	at Brent \$70/bbl	at Brent \$100/bbl
0.031	3.15%	$\sim 3\%$	$\pm \$2.20/\text{bbl}$	$\pm \$3.10/\text{bbl}$
0.059	6.08%	$\sim 6\%$	$\pm \$4.25/\text{bbl}$	$\pm \$6.10/\text{bbl}$
0.075	7.79%	$\sim 8\%$	$\pm \$5.45/\text{bbl}$	$\pm \$7.80/\text{bbl}$
0.109	11.52%	$\sim 12\%$	$\pm \$8.10/\text{bbl}$	$\pm \$11.50/\text{bbl}$
0.127	13.55%	$\sim 14\%$	$\pm \$9.50/\text{bbl}$	$\pm \$13.50/\text{bbl}$

Reading the slide table with this:

- Convex SCM Hormuz train 0.059 $\rightarrow \sim 6\%$ typical pre-period error ($\sim \$4\text{--}5/\text{bbl}$ at Hormuz prices).
- Convex SCM Russia val 0.127 $\rightarrow \sim 14\%$ val error ($\sim \$9\text{--}11/\text{bbl}$ at Russia prices). The widest fit gap on the board, reflecting that the simplex constraint cannot span Brent’s 2020–22 range with the audited donor set.
- Bayesian Ridge train 0.031 / val 0.096 $\rightarrow \sim 3\%$ in training but $\sim 10\%$ in val. The gap between training-period precision and val-period error is the overfit signature.

Behind the slide

The synthetic represents “what Brent would have done given how the donors moved.” If donors moved a lot (Russia post-period) the synthetic naturally moves too; sparse models drop the macro-driven donors and the synthetic stays flatter. BR keeps them in and the synthetic rises with Brent. The model spread on Russia (0.5% to 36.8%) is itself diagnostic — it tells you the macro confounder is generating model-specific identification problems, not just numerical noise.

If asked

- *Why don’t you exclude the overfit models?* The heuristic threshold (val/train > 2) is just a flag, not a gate. Auto-excluding would impose a literature-unsupported cutoff and would tighten the IQR artificially. Defense is the IQR-as-uncertainty-band convention.
- *Is the validation downward biased by hyperparameter selection?* Yes; documented. Magnitude: the bias is small ($\sim 10\%$ on val RMSE) because grids are tight per Cawley & Talbot.
- *Doesn’t the 2022 macro confounder invalidate the Russia SCM?* Weakens, doesn’t fully invalidate. Identification claim becomes “Russia premium + asymmetric macro impact” rather than pure Russia ATT. Defended by (a) EIA STEO agreement within $\$3.98/\text{bbl}$, (b) convergence of sparse models 22–30%, (c) historical record $\$95 \rightarrow \130 , (d) Hormuz works cleanly as the methodology validation. We use Russia as a magnitude-credibility check, not as a clean causal ATT.
- *Why is RMSE in log-Brent and not absolute Brent?* Standard SCM practice (Abadie 2010 fits log-levels). Log scale makes the relative error comparable across price regimes (a $\$5/\text{bbl}$ miss at $\$30$ is much worse than at $\$120$). Also makes the SCM gap interpretable as a percentage directly.
- *Why log-levels and not log-returns?* SCM is about counterfactual *paths* (where would Brent’s level be), not about return prediction. Fitting returns and then integrating compounds errors; fitting levels directly produces the path you want to report.

12. Slide 12 — Validation results: in-space placebo

[90s]

Talking points

- **Plain-English summary:** re-run the same SCM machinery 21 more times, each time pretending a different donor was the treated unit instead of Brent. If Brent’s gap is bigger than every fake gap the procedure invents for these untreated donors, the effect is unlikely to be something the model would have produced by accident.
- Abadie’s randomization inference: refit treating each donor as the placebo treated unit, compute Brent’s rank in the post/pre RMSPE distribution.
- **Permutation floor:** $1/22 \approx 0.045$ for 21-donor pool; $1/34 \approx 0.029$ for 33-donor full pool.
- Hormuz: **all 5 models hit the floor under both pools** — Brent ranks 1 of 22 in each. Strongest possible permutation signal.
- Russia: not decisively rejected. Only XGBoost reaches floor at 21-donor (0.045) and slips to 0.147 at 33-donor. Other models $p \sim 0.4\text{--}0.7$.

Behind the slide

The two-pool design (21-donor and 33-donor full) is a robustness check. The 21-donor floor is too coarse to distinguish “Brent is rank 1” from anything below 0.045. The 33-donor pool with audit-flagged donors mixed back in tests whether the rank-1 finding holds when “noisier” donors expand the placebo distribution. Hormuz is robust; Russia is not.

The Russia weakness is the known limitation of in-space placebo at pre-periods with high donor volatility (Russia’s pre-period 2020-07 $\rightarrow$ 2022-02 spans COVID recovery + 2021 reflation rally, both of which moved donors). This widens the placebo distribution and prevents Brent’s post-period ratio from sitting in the tail.

If asked

- *Is XGBoost’s Russia rejection meaningful?* It clears at 21-donor floor but fails at 33-donor ($p \sim 0.147$), so the rejection is fragile. Russia’s case for a real effect rests on (a) the historical record (well-documented \$95 $\rightarrow$ \$130 move), (b) the audit-supported SUTVA defense, and (c) the cross-event weight transfer (slide A5), not on the in-space placebo alone.
- *Why does Hormuz have such tight p-values?* Brent’s post/pre RMSPE ratios for Hormuz are 6 to 12 across models — much bigger than any donor’s. The donor pool is also more stable in the Hormuz pre-window (no COVID-style regime shifts), so the placebo distribution is tight.
- *Permutation test theory: how do we know p-values are valid?* The randomization-inference framework (Fisher 1935; Lehmann & Romano 2005) is what justifies SCM’s p-values. Abadie 2010 §4 explicitly grounds the placebo inference in this framework. Exact at the unit level (we enumerate all 22 or 34 placebo refits).

13. Slide 13 — External validation: EIA STEO pre-invasion forecasts (Russia only)

[75s]

Talking points

- Two-method cross-check: SCM Russia counterfactual vs EIA STEO’s pre-invasion forecast of 2022 Brent, measured on the *exact same 151-day treatment window* (Feb 24 $\rightarrow$ Sep 30 2022).
- Two STEO vintages: **Jan-22** (issued $\sim$ Jan 11, 2022 — 6 weeks pre-invasion) and **Feb-22** (issued $\sim$ Feb 8, 2022 — 16 days pre-invasion, last pre-invasion STEO).
- Result table: SCM ensemble-median counterfactual \$88.92/bbl. STEO Feb-22 step-function \$84.94/bbl. **Disagreement: only \$3.98/bbl, or 5.7 pp on implied ATT.**
- Three of five SCM models bracket STEO Feb-22 within $\pm$ \$6.

- Implied ATTs: SCM-based 22.1%, STEO-Feb-22-based 27.8% ($\Delta \sim 5.7$ pp), STEO-Jan-22-based 43.4% (further in time = larger gap).

Behind the slide

SCM and STEO share no information path: SCM uses cross-asset co-movement; STEO uses EIA's structural supply/demand model. The agreement is independent-method evidence the SCM isn't producing a fantasy counterfactual. The disagreement direction matters too: SCM is more *conservative* (lower implied ATT) because donor co-movement absorbs macro spillover (inflation, demand recovery) that the STEO doesn't.

This is uniquely available for Russia: Hormuz has no equivalent pre-event structural forecast because Hormuz happened in early 2026 and the methodology was developed after the event.

If asked

- *Why use STEO and not other forecasts?* STEO is the most prominent monthly oil forecast and publicly available with archive vintages. EIA Short-Term Energy Outlook archives at eia.gov/outlooks/steo/archives.
- *Is STEO the ground truth?* No; STEO is another model. The agreement says two independent methods produce the same counterfactual, not that either is correct.
- *What if STEO is just wrong?* Then the SCM might also be wrong in the same direction. The two-method agreement is necessary but not sufficient evidence. The qualitative case rests on the broader battery (placebo + cross-event transfer + donor audit).

14. Slide 14 — Headline result: ATT estimate

[90s]

Talking points

- **Headline numbers:**
 - Hormuz 2026 ensemble median: **$\sim 43\text{--}44\%$** chokepoint premium over the 3-month post-window. IQR [40.3, 44.2].
 - Russia 2022 ensemble median: **$\sim 21\text{--}22\%$** drift-adjusted Brent premium over the 7-month post-window. IQR [13.1, 23.8].
- Hormuz IQR is tighter than Russia (4 pp vs 11 pp) — donors are more stable in the Hormuz pre-period.
- Drift adjustment: subtract the pre-period slope $\times$ post-window length per Roth 2022 (pretest-with-caution framework).
- Hormuz: validated by in-space placebo at the floor under both pools across all 5 models, plus cross-event weight transfer (appendix). Russia: matches EIA STEO to within \$3.98/bbl.

Behind the slide

This is the punchline of Part 1. The Hormuz $\sim 43\%$ is the substantive headline; Russia $\sim 22\%$ is the methodology-validation case. The cross-event weight transfer (appendix) shows that the same factor structure that fits Russia transfers to Hormuz for the linear/sparse models (Convex SCM, ASCM, Elastic-net), giving regime-stability evidence for the focal estimate.

If asked for a single number for Hormuz: 43%. If asked for a range: 38–50% across models, 40–44% interquartile.

If asked

- *How does 43% compare to scenario-model estimates?* Dallas Fed WP 2609, Kiel Policy Brief 206, and IEA OMR April–May 2026 report scenario-model estimates for partial vs full Hormuz closure. The SCM is in the same order of magnitude as the partial-closure scenarios.
- *What's the standard error?* The IQR is the model-uncertainty band; no formal SE is reported because SCM doesn't naturally have one. Inference is via the in-space placebo p-value.

- *Is the Hormuz post-window long enough?* 59 obs is short by SCM convention. The cross-event weight transfer substitutes inferential depth.

15. Slide 15 — Part 2: future directions

[45s]

Talking points

- Three directions build directly on the Brent SCM result established in Part 1. State each in one sentence; do not over-commit to any.
- **Direction 1 — LNG market causal estimation (JKM, TTF).** Change the treated asset rather than the methodology: apply the same SCM design to JKM Asian LNG spot and/or TTF European LNG spot. Identifies the Hormuz LNG premium paid by Asian buyers vs a cross-asset counterfactual. *First causal estimate of the Hormuz effect on Asian LNG pricing.* Same machinery, different treated asset.
- **Direction 2 — Local projections (LP) across horizons.** Estimate the Brent treatment effect at $h = 1, 5, 14, 30$ days. Produces a horizon-by-horizon impulse response: how fast the effect builds and how long it persists. SCM gives the *average* ATT over a window; LP gives the *shape* of the path — different and complementary information.
- **Direction 3 — Retail fuel and CPI pass-through.** Estimate what share of the $\sim 44\%$ Brent premium passed through to retail gasoline and headline CPI in oil-importing economies. Converts the upstream ATT into a consumer-price impact estimate. The Brent ATT is the upstream input; pass-through delivers the policy-relevant inflation number.
- Closing (red box): we are currently considering these three options, and are open to discussing further options/suggestions.

Behind the slide

The Part 2 slide is intentionally a menu rather than a commitment. Present the three directions evenly; if asked to rank, the LNG-spot extension is the most natural next step because it reuses the entire SCM machinery with a different treated asset. The LP extension complements Part 1's average ATT with a horizon-decomposed impulse response. The pass-through extension is the most policy-relevant but adds an entirely new dataset (retail prices, CPI) and identification challenge (incomplete pass-through, country heterogeneity).

If asked

- *Which direction would you actually pick?* Honest answer: LNG-spot SCM is the easiest path to a clean second result (same methodology, new treated asset). LP is the most rigorous extension methodologically. Pass-through is the most policy-impactful but requires more data engineering.
- *Could you do all three?* Time-budget concern; one is realistic for the thesis timeline. Multiple would dilute depth.
- *Would the network framing reappear in Part 2?* Possible: maritime-routing features could enter the donor pool, or a separate propagation model could layer on top of the ATT. Not in the current shortlist but open to suggestion.

16. Slide 20 — Appendix A3: in-time placebo and LOO

This is reserve material; do not budget time. Use only if asked about robustness of the headline. The slide shows two diagnostic tests side-by-side — both refit the SCM under a deliberate perturbation and report how the headline ATT moves.

Procedure: in-time placebo (Abadie, Diamond & Hainmueller 2015)

Pretend the treatment happened 6 months *earlier* than the actual T_0 , with everything else held identical. If the SCM produces a large “effect” at the fake date — where we know nothing happened — the procedure is fragile.

Formula.

$$T_0^{\text{fake}} = T_0 - 6 \text{ months}$$

(re-fit SCM with pre-period truncated at T_0^{fake} ; default hparams, no re-tuning)

$$\text{gap}_t^{\text{fake}} = y_t - \hat{y}_t^{\text{fake}} \quad \text{for } t \in [T_0^{\text{fake}}, T_0)$$

$$\text{in-time gap (\%)} = 100 \cdot (\exp(\overline{\text{gap}_t^{\text{fake}}}) - 1)$$

where $\overline{\text{gap}_t^{\text{fake}}}$ is the mean log-gap over the fake post-period (the stretch of pre-event data we're pretending is post-event).

Unit: **percent** — the same unit as the headline ATT.

Reading the numbers.

- **Hormuz** fake window: late 2025 → Feb 2026. In-time gaps are -10 to -12% across all 5 models, all in the same direction. Interpretation: the donor pool was running slightly hotter than Brent in late 2025, so any SCM fit produces a small *negative* placebo gap. This is not a model failure — it tells us the pre-Hormuz runup was concentrated in donors, not Brent.
- **Russia** fake window: Aug 2021 → Feb 2022. In-time gaps span -5% to $+26\%$ with mixed signs. This window contains both the late-2021 oil rally and the Jan-2022 invasion runup, so partial real-effect contamination is *expected*. The XGBoost $+26\%$ is low-power (training period shrinks $\sim 31\%$ when 6 months are sliced off).
- Rule of thumb: a fake-period gap with absolute value $\lesssim$ the pre-period noise floor (a few %) is reassuring; a fake-period gap comparable in magnitude to the real ATT would be alarming.

Procedure: leave-one-donor-out (LOO; Abadie, Diamond & Hainmueller 2015)

Sensitivity check: re-fit the SCM with each “weight-bearing” donor dropped in turn, and look at how much the headline ATT moves across the drops. If one donor is silently carrying the whole estimate, the SCM is fragile.

Formula. For each donor j with non-trivial weight (e.g. $w_j > 0.05$, or top- $K = 5$ for nonlinear models):

$$\text{pool}^{(-j)} = \text{donors} \setminus \{j\} \quad (20\text{-donor pool with } j \text{ removed})$$

$$\text{ATT}^{(-j)} = 100 \cdot (\exp(\overline{\text{gap}_t^{(-j)}}) - 1) \quad \text{re-fit SCM on the 20-donor pool}$$

$$\text{LOO range (pp)} = \max_j \text{ATT}^{(-j)} - \min_j \text{ATT}^{(-j)}$$

where the max and min are taken across the leave-out runs (typically 5–10 per model, depending on how many donors carry weight).

Unit: **percentage points (pp)** — the spread of the leave-one-out ATT distribution. *Not* a percent. “5 pp” means the ATT could be 5 percentage points lower or higher depending on which donor you drop, e.g. a 40% headline could become anywhere in $[37.5\%, 42.5\%]$ under leave-one-out.

Reading the numbers.

- **Tightest (good): XGBoost** at ~ 5 pp for both events. No single donor's removal moves the ATT meaningfully — the result is robust to the donor pool composition.
- **Widest (bad): Bayesian Ridge Russia** at 24.9 pp, with the † flag indicating the range *crosses zero* (min -12.8% to max $+12.0\%$). The headline can flip sign just by dropping one donor — the model is LOO-fragile, consistent with its overfit signature (walk-forward val/train ratio 3.07) and degenerate $+0.5\%$ baseline ATT.

- Hormuz LOO ranges are mostly 5–13 pp across models, narrower than Russia. The ensemble median is robust because four of five models (Convex SCM, ASCM, Elastic-net, XGBoost) agree closely.
- Rule of thumb: LOO range < 10 pp = robust; 10–20 pp = sensitive but acceptable; > 20 pp or crosses zero = fragile, flag for narrative discussion.

If asked

- *Why 6 months for the in-time placebo offset?* Standard convention from Abadie 2015; long enough that any fake gap is not contaminated by anticipation of the real T_0 , short enough not to halve the training window. Six months is the typical choice in applied SCM papers.
- *Why not bootstrap the LOO range for a CI?* The LOO range is already a robustness statistic, not an SE; it captures donor-composition sensitivity rather than sampling variability. The in-space placebo (slide 12) handles the sampling-style inference question.
- *Why does Bayesian Ridge fail LOO so badly?* BR is the only dense (no-feature-selection) model in the ensemble; dropping any one donor reshuffles the entire coefficient vector. Sparse models (Convex SCM, Elastic-net) pin to a small donor set and stay stable.

17. Slide 16 — Thank you / Q&A

Open the floor. The five appendix slides (17–21) are reserve material; navigate to them on demand:

- **Slide 17 — Appendix: excluded donors.** 12 Russia-treated drops with audit reasons; categorical exclusions (energy complex, oil-currencies). Lists Russia H-tier (Wheat, Corn, Palladium, EUR, DXY, BTC) and M-tier (Copper, IronOre, Soybeans, EM_Eq, GBP, US10Y).
- **Slide 18 — Appendix: donor importance.** Per-model donor importance heatmap, normalised within each column to sum to 1 (so colour scale is comparable across models). Convex weights for SCM/ASCM; $|\hat{\beta}_j|$ for Elastic-net/Bayesian Ridge; feature importance (gain-based) for XGBoost.
- **Slide 19 — Appendix: donor SUTVA cleanliness.** Permutation mean-shift test results (Chow 1960 hypothesis + Lehmann & Romano 2005 permutation + Phipson & Smyth 2010 correction). Hormuz 1/33 flagged (IronOre); Russia 1/33 (CNY). Audit agreement: Hormuz 32/33, Russia 20/33.
- **Slide 20 — Appendix: in-time placebo & LOO.** Combined diagnostics table. See dedicated section above for the formulas, units, and reading. Headline: Hormuz in-time gaps –10 to –12% uniform across models (background donor drift, not model failure); Russia in-time gaps mixed sign (fake window contains pre-invasion runup, contamination expected). LOO ranges 5–13 pp for Hormuz, 5–25 pp for Russia; Bayesian Ridge Russia is the LOO-fragile case (24.9 pp, range crosses zero).
- **Slide 21 — Appendix: cross-event weight transfer.** Russia-fitted weights applied to Hormuz panel: Convex SCM transfers cleanly (+8 pp drift), ASCM and Elastic-net transfer with magnitude attenuation, XGBoost and Bayesian Ridge transfer pathologically (transferred pre-RMSPE 22× and 314× independent). The relevant subtlety: per-donor weights are not literally stable across events (JPY moves from rank 12 to rank 1 in convex SCM), but the *projection onto donor space* is. Cite Abadie & L’Hour 2021 on convex SCM weight non-uniqueness.

General Q&A backup material

Methodological choices to be ready to defend

- **Why 5 models and not 1?** Ensemble spans the SCM regularization design space; no single model assumption is privileged; IQR is the honest uncertainty band.
- **Why log-Brent and log-donors?** Stationarity (log-returns are approximately stationary; levels are not), interpretability (log differences $\approx$ percentage changes), variance stabilization.

- **Why a shared 21-donor pool for both events?** Enables the cross-event weight transfer test (the strongest single methodology generalization test). Alternative per-event pools would give somewhat different headline numbers but break the cross-event design.
- **Why Russia 2022 specifically as the validation case?** Most recent large oil shock with a long post-window; independent of Hormuz pre-period; well-documented; matches the Hormuz design template.
- **Why pre-windows starting 2020-07 (Russia) and 2024-06 (Hormuz)?** 18–20 month pre-windows balance sample size against regime stability. Earlier starts include too much COVID disruption (Russia) or pre-2024 macro instability (Hormuz). Documented in PRE_WINDOWS at lib/config.py:29.

Numbers to know cold

- Hormuz headline drift-adjusted ATT: **43%** (IQR 40–44).
- Russia headline drift-adjusted ATT: **21–22%** (IQR 13–24).
- Actual Russia Brent mean over Feb 24 → Sep 30 2022: \$108.54/bbl.
- SCM ensemble median Russia counterfactual: \$88.92/bbl.
- EIA STEO Feb-22 forecast for the same window: \$84.94/bbl.
- Hormuz pre-event window: 423 obs / Russia pre-event: 421 obs.
- Hormuz post-event obs: ~59 (the binding constraint on in-event inference).
- In-space placebo floors: $1/22 = 0.045$ (21-donor); $1/34 = 0.029$ (33-donor).

Hardest critiques

- **“Your Hormuz IQR is 4 pp — too narrow given 60 obs.”** The IQR reflects model agreement, not statistical uncertainty. The structural uncertainty is in the in-space placebo (floor) and cross-event transfer (degraded RMSPE for nonlinear models). Honest reading: 38–50% across all models, 40–44% across the agreeable models.
- **“Cross-event weight transfer fails for 2 of 5 models.”** It fails for the dense / regime-dependent models (XGBoost, Bayesian Ridge) and works for the sparse models (Convex SCM, ASCM, Elastic-net). This is mechanically expected: sparse weights pin to factor-stable donors. The Hormuz headline rests on the sparse models, which transfer.
- **“In-space placebo for Russia is weak.”** Yes. Russia’s defense is multi-layered: EIA STEO external validation, historical record (\$95 → \$130 move is documented), audit-supported SUTVA, cross-event weight transfer. The in-space placebo is one of four pieces of evidence.
- **“You changed the in-time placebo to use defaults instead of tuned hparams.”** Yes, intentionally: the fake-post period overlaps the val window used for hparam tuning, so tuned hparams would induce second-order leakage. Defaults are the literature-defensible choice for an unselection-bias test. Documented in validation.md §5e (ii).

Companion compiled from docs/methodology.md, docs/validation.md, docs/results.md, docs/donor_catalog.md, and docs/motivation.md as of commit 8721f35. For full details on any item, the source-of-truth documents are in docs/.